z-transform and ROC

Answers and worked solutions · Day 13 · Thu 5 Nov · pack D7

5. Practice

From your course files

T6 Q1 (tutorial).

Hint: $2^n u(-n-1)$ has a minus sign in its transform, and (a) and (b) have the same algebraic $X(z)$. Only the ROC tells them apart.

(a) $\left(\frac{1}{3}\right)^n u(n) \leftrightarrow \dfrac{1}{1-\frac{1}{3}z^{-1}}$, $|z| > \frac{1}{3}$, and $2^n u(-n-1) \leftrightarrow -\dfrac{1}{1-2z^{-1}}$, $|z| < 2$. So

$$X_1(z) = \frac{1}{1-\frac{1}{3}z^{-1}} - \frac{1}{1-2z^{-1}} = -\frac{\frac{5}{3}z^{-1}}{1-\frac{7}{3}z^{-1}+\frac{2}{3}z^{-2}}, \qquad \text{ROC: } |z| > \tfrac{1}{3} \cap |z| < 2 = \tfrac{1}{3} < |z| < 2.$$

Sketch: a ring between the pole circles at $\frac{1}{3}$ and 2 (it contains the unit circle).

(b) Both pieces are causal: $\left(\frac{1}{3}\right)^n u(n) - 2^n u(n) \leftrightarrow \dfrac{1}{1-\frac{1}{3}z^{-1}} - \dfrac{1}{1-2z^{-1}}$, so

$X_2(z) = -\dfrac{\frac{5}{3}z^{-1}}{1-\frac{7}{3}z^{-1}+\frac{2}{3}z^{-2}}$ (the same as X_1), with ROC $|z| > \frac{1}{3} \cap |z| > 2 = |z| > 2$.

Sketch: everything outside the circle of radius 2.

(c) Time shift: $x(n+4) \leftrightarrow z^4 X(z)$.

$$X_3(z) = z^4 X_1(z) = -\frac{\frac{5}{3}z^{3}}{1-\frac{7}{3}z^{-1}+\frac{2}{3}z^{-2}}, \qquad \text{ROC: } \tfrac{1}{3} < |z| < 2.$$

The ring already excludes $z = \infty$, so the shift does not change the ROC.

(d) Time reversal: $x(-n) \leftrightarrow X(z^{-1})$, and the ROC radii are inverted.

$$X_4(z) = X_1(z^{-1}) = -\frac{\frac{5}{3}z}{1-\frac{7}{3}z+\frac{2}{3}z^{2}}, \qquad \text{ROC: } \tfrac{1}{2} < |z| < 3.$$

The poles move from $\frac{1}{3}, 2$ to $3, \frac{1}{2}$.

T6 Q2 (tutorial).

Hint: the ROC of Y is the ring $\cap\, |z| > \frac{1}{2}$. The partial-fraction step is D8 material.

$y(n) = x_1(n) * x_2(n)$, so $Y(z) = X_1(z)X_2(z)$ with ROC $R_{x_1} \cap R_{x_2}$. From Q1(a), $X_1(z) = -\dfrac{\frac{5}{3}z^{-1}}{1 - \frac{7}{3}z^{-1} + \frac{2}{3}z^{-2}}$, $\frac{1}{3} < |z| < 2$, and $X_2(z) = \dfrac{1}{1 - \frac{1}{2}z^{-1}}$, $|z| > \frac{1}{2}$.

$$Y(z) = -\frac{\frac{5}{3}z^{-1}}{\left(1 - \frac{1}{3}z^{-1}\right)\left(1 - 2z^{-1}\right)\left(1 - \frac{1}{2}z^{-1}\right)}, \qquad \text{ROC: } \frac{1}{2} < |z| < 2.$$

Partial fractions (cover-up):

$$Y(z) = -\frac{2}{1 - \frac{1}{3}z^{-1}} - \frac{\frac{4}{3}}{1 - 2z^{-1}} + \frac{\frac{10}{3}}{1 - \frac{1}{2}z^{-1}}.$$

In the ring $\frac{1}{2} < |z| < 2$, the poles at $\frac{1}{3}$ and $\frac{1}{2}$ are inside (causal terms) and the pole at 2 is outside (anti-causal term, $-\frac{1}{1-2z^{-1}} \leftrightarrow 2^n u(-n-1)$):

$$y(n) = -2\left(\frac{1}{3}\right)^n u(n) + \frac{4}{3}\,2^n u(-n-1) + \frac{10}{3}\left(\frac{1}{2}\right)^n u(n).$$

T6 Q3 (tutorial).

Hint: the figure is h_1 in cascade with (h_2 minus the $h_3 \to h_4$ branch). $H_2 = H_3$ lets you factor $H_2(1 - H_4)$.

Transfer functions of the blocks:

$$H_1(z) = \tfrac{1}{2} + \tfrac{1}{4}z^{-1} + \tfrac{1}{2}z^{-2}, \qquad H_2(z) = H_3(z) = 1 - z^{-1}, \qquad H_4(z) = z + 2z^{-2}.$$

Overall system:

$$H(z) = H_1(z)\big[H_2(z) - H_3(z)H_4(z)\big] = H_1(z)H_2(z)\big[1 - H_4(z)\big] = \left(\tfrac{1}{2} + \tfrac{1}{4}z^{-1} + \tfrac{1}{2}z^{-2}\right)\left(1 - z^{-1}\right)\left(-z + 1 - 2z^{-2}\right)$$

$$H(z) = -\tfrac{1}{2}z + \tfrac{3}{4} - \tfrac{1}{2}z^{-1} - \tfrac{1}{4}z^{-2} - \tfrac{1}{2}z^{-4} + z^{-5}.$$

Read the coefficients off as the impulse response (the z^{+1} term is at $n = -1$):

$$h(n) = \Big\{ -\tfrac{1}{2}, \underset{\uparrow}{\tfrac{3}{4}}, -\tfrac{1}{2}, -\tfrac{1}{4}, 0, -\tfrac{1}{2}, 1 \Big\}.$$

The system is **not causal**, because $h(-1) = -\frac{1}{2} \neq 0$ (that is, $h(n) \neq 0$ for $n < 0$).

All answers above were re-derived in Python (sympy) and agree with the official solutions.

New questions

N1.

a. $X(z) = 1 - 2z^{-1} + z^{-2} = (1 - z^{-1})^2$. Only positive n appear, so **ROC:** $|z| > 0$ (all z except 0).

b. $X(z) = \dfrac{(z-1)^2}{z^2}$. A double zero at $z = 1$ and a double pole at $z = 0$. $X(\infty) = 1$, so nothing at ∞. Count 2 = 2.

c. $x(n+2) \leftrightarrow z^2 X(z) = z^2 - 2z + 1$. Samples are now at $n = -2, -1, 0$, so **ROC:** $|z| < \infty$ (all z except ∞). The shift removed the exclusion at 0 and added one at ∞.

N2.

Euler: $\cos(\pi n/2) = \frac{1}{2}\left(e^{j\pi n/2} + e^{-j\pi n/2}\right)$, so $x(n) = \frac{1}{2}\left(\frac{j}{2}\right)^n u(n) + \frac{1}{2}\left(-\frac{j}{2}\right)^n u(n)$.

$$X(z) = \frac{1}{2}\left[\frac{1}{1 - \frac{j}{2}z^{-1}} + \frac{1}{1 + \frac{j}{2}z^{-1}}\right] = \frac{1}{1 + \frac{1}{4}z^{-2}} = \frac{z^2}{z^2 + \frac{1}{4}}$$

ROC: $|z| > \frac{1}{2}$ (both pieces causal, $|\pm j/2| = \frac{1}{2}$).

Poles at $z = \pm j\frac{1}{2}$. A double zero at $z = 0$.

Check: $x(n) = \{1, 0, -\frac{1}{4}, 0, \frac{1}{16}, \ldots\}$, and the series of $1/(1 + \frac{1}{4}z^{-2})$ is $1 - \frac{1}{4}z^{-2} + \frac{1}{16}z^{-4} - \ldots$. This agrees. (Same as the table row with $a = \frac{1}{2}$, $\cos\omega_0 = 0$.)

N3.

a. $\dfrac{z^{-3}}{1 - \frac{1}{2}z^{-1}}$, $|z| > \frac{1}{2}$. Still causal, so ∞ stays in the ROC.

b. $\dfrac{1}{1 - \frac{1}{2}z}$, $|z| < 2$.

c. $X(z/2) = \dfrac{1}{1 - z^{-1}}$, $|z| > 1$. This is $u(n)$, as it should be: $2^n(\frac{1}{2})^n = 1$.

d. $\dfrac{\frac{1}{2}z^{-1}}{(1 - \frac{1}{2}z^{-1})^2}$, $|z| > \frac{1}{2}$.

e. $X(z^2) = \dfrac{1}{1 - \frac{1}{2}z^{-2}}$, $|z| > \left(\frac{1}{2}\right)^{1/2} = 0.707$. The poles are now at ± 0.707.

f. $x(0) = \lim_{z\to\infty} X(z) = \mathbf{1}$. This agrees with $(\frac{1}{2})^0 = 1$.

N4.

a. $X(z) = \dfrac{1}{1-\frac{1}{4}z^{-1}}$, $|z| > \frac{1}{4}$. $H(z) = z - \frac{1}{4} = z\left(1 - \frac{1}{4}z^{-1}\right)$, $|z| < \infty$.

b. $Y(z) = z\left(1 - \frac{1}{4}z^{-1}\right) \cdot \dfrac{1}{1-\frac{1}{4}z^{-1}} = z$. ROC: $|z| < \infty$ (the whole plane except ∞). So $\mathbf{y(n) = \delta(n+1)}$.

c. **Not causal**: $h(-1) = 1 \neq 0$ (the z^{+1} term). $R_x \cap R_h = \frac{1}{4} < |z| < \infty$, but the zero of H at $z = \frac{1}{4}$ cancels the pole of X. With no pole left at $\frac{1}{4}$, the ROC grows to include $|z| \leq \frac{1}{4}$. This is the "at least" in the convolution property.

N5.

a. Loop: $H_L = \dfrac{G}{1+GF} = \dfrac{z^{-1}}{1-0.8z^{-1}}$. Cascade with $H_1(z) = 1 + z^{-1}$:

$$H(z) = \frac{z^{-1}(1+z^{-1})}{1-0.8z^{-1}} = \frac{z+1}{z(z-0.8)}$$

b. Zero at $z = -1$. Poles at $z = 0$ and $z = 0.8$. $H(\infty) = 0$, so there is a zero at $z = \infty$. Count: 2 zeros and 2 poles.

c. All blocks are causal and the loop has a delay, so the system is **causal** with **ROC** $|z| > 0.8$.

d. The ROC contains $|z| = 1$, so $H(\omega)$ exists. $H(1) = \dfrac{2}{0.2} = \mathbf{10}$. $H(-1) = \dfrac{0}{\dots} = \mathbf{0}$ (the zero on the unit circle). Low-pass.

N6.

a. Split: $a^n u(n)$ for $n \geq 0$ gives $\dfrac{1}{1-az^{-1}}$, $|z| > |a|$. For $n \leq -1$, $a^{-n}u(-n-1) = \sum_{m\geq 1} a^m z^m = \dfrac{az}{1-az}$, $|z| < \frac{1}{|a|}$.

$$X(z) = \frac{1}{1-az^{-1}} + \frac{az}{1-az} = \frac{(1-a^2)z}{(z-a)(1-az)}, \qquad |a| < |z| < \frac{1}{|a|}$$

The ring is non-empty only if $|a| < 1/|a|$, which means $|a| < 1$.

b. $a = \frac{1}{2}$: $X(z) = \dfrac{\frac{3}{4}z}{(z-\frac{1}{2})(1-\frac{1}{2}z)} = \dfrac{-3z}{(2z-1)(z-2)}$. Poles at $\frac{1}{2}$ and 2. A zero at 0, and a zero at ∞ ($X(\infty) = 0$). ROC: $\frac{1}{2} < |z| < 2$. This is the same annulus as the figure in Example 2.

c. $X(0) = X(1) = \dfrac{-3}{(1)(-1)} = \mathbf{3}$. Check: $\sum(\frac{1}{2})^{|n|} = 2 + 1 = 3$.
$X(\pi) = X(-1) = \dfrac{3}{(-3)(-3)} = \mathbf{\frac{1}{3}}$.

d. $a = 2$: the pieces need $|z| > 2$ and $|z| < \frac{1}{2}$. These do not overlap, so the ROC is empty and **no z-transform exists**.

6. Quick recall check

Question

1. The exterior of a circle, $|z| > r$, through the outermost pole (it includes ∞).
2. $z^3 X(z)$.
3. $0.25 < |z| < 5$.
4. On the unit circle $z = e^{j\omega}$. It exists only if the ROC contains $|z| = 1$.
5. No. The z^{+1} term means $h(-1) = 1 \neq 0$.